

Pranav Puppala

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EDUCATION

The University of Texas at Dallas

Master of Science, Computer Science - Intelligent Systems

Richardson, TX

Aug. 2026 – Dec. 2027

The University of Texas at Dallas

Bachelor of Science, Computer Science, **GPA: 3.72**

Richardson, TX

Jan. 2023 – May 2026

- **Relevant Coursework:** Data Structures and Algorithms, Machine Learning, Data Science, Operating Systems, Databases, Automata Theory, Advanced Algorithms

EXPERIENCE

Software Engineer Intern (Capstone Project)

UT Southwestern Medical Center

Feb. 2026 – May 2026

Dallas, TX

- Built a **hybrid RAG pipeline** using vector retrieval and cross-encoder reranking to generate persona-aware patient simulation responses for medical training
- Designed a **synthetic persona generation pipeline** that extracts structured clinical profiles from conversational transcripts to enable real-time simulation
- Developed a **real-time patient simulation system** with structured behavioral metadata tagging to output natural dialogue and structured JSON schema

PROJECTS

EDA Agent | Django REST, React, LangGraph, Groq, Pandas

Aug. 2026 – Sep. 2026

- Built an **autonomous EDA agent** using LangGraph and Groq that accepts CSV uploads, investigates datasets across up to 7 self-directed loop iterations, and generates markdown reports with embedded visualizations
- Architected a separation between **8 deterministic analysis tools** (pandas/numpy) and LLM orchestration nodes, enforcing one tool call per iteration with structured Pydantic output for reliable loop termination
- Developed a **full-stack interface** with a Django REST backend and React/Tailwind frontend featuring per-iteration agent reasoning traces and a rendered markdown report viewer

KnowledgeAtlas | Django REST, React, LangChain, Celery, Docker

Jun. 2026 – Jul. 2026

- Built a full-stack **RAG application** with a Django REST backend and React frontend, using LangChain and Groq to generate **streamed, citation-backed answers** grounded in retrieved source content
- Architected an **asynchronous ingestion pipeline** with Celery and Redis, containerizing the full stack with Docker Compose for one-command deployment
- Designed a vector retrieval system with ChromaDB, **calibrating a similarity threshold against real test data** and building an evaluation harness with a **100% pass rate** across curated queries

ExpenseTracker | Django REST, React, JWT, Axios, Recharts, Tailwind CSS

Aug. 2025 – Oct. 2025

- Engineered a **full-stack expense tracking application** using Django REST Framework and React with JWT authentication, token rotation, and silent refresh via Axios interceptors across **8 protected API endpoints**
- Architected a service layer across **16 REST API endpoints**, implementing ownership-scoped queriesets, cursor-based pagination, and multi-parameter filtering by keyword, category, and date range
- Built an **interactive React dashboard** with Recharts displaying a zero-filled 30-day trend line, monthly analytics, and category breakdowns styled with Tailwind CSS and shadcn/ui

TECHNICAL SKILLS

Languages: Java, Python, C++, SQL, PostgreSQL, JavaScript, HTML, CSS

Frameworks: React, Next.js, Django, Django REST Framework, Scikit-Learn, TensorFlow

Developer Tools: Git, VS Code, GitHub, Docker, Claude Code, Cursor

Libraries: Pandas, NumPy, Matplotlib, Axios, Tailwind CSS, LangChain, LangGraph, ChromaDB, Celery, Redis